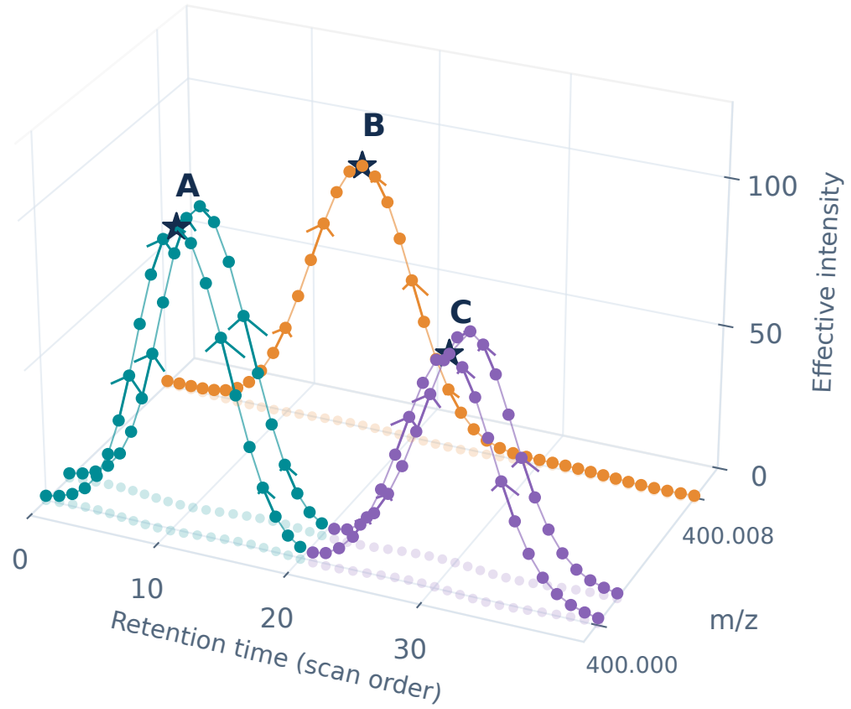
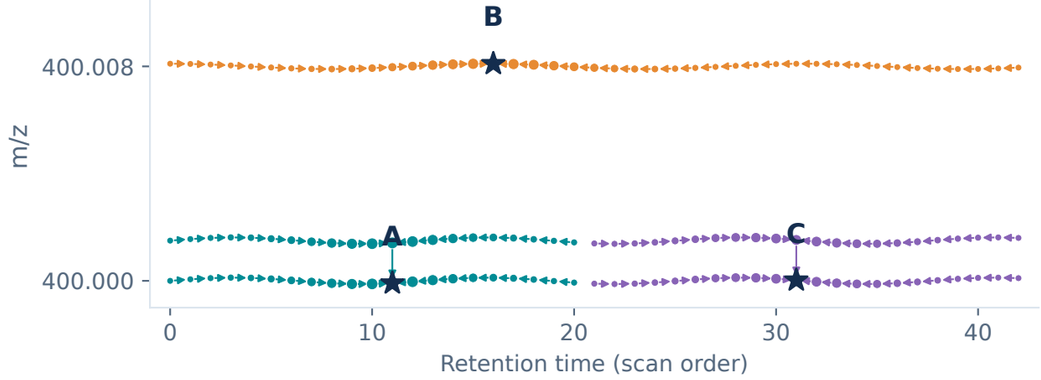
What does the 3D in Peak3D mean?

An intensity landscape over retention time and m/z, represented by connected centroids.



A and C: same m/z, different RT. B: a co-eluting ion at a different m/z.

The same graph, viewed from above



Position: retention time + m/z

Links use nearby scans and pair-specific mass tolerance.

Height: effective intensity

Same-scan split pieces contribute to the local height.

Colour: centroids assigned to the same surviving apex.

Time and m/z define local neighbours; intensity directs their paths toward apexes.

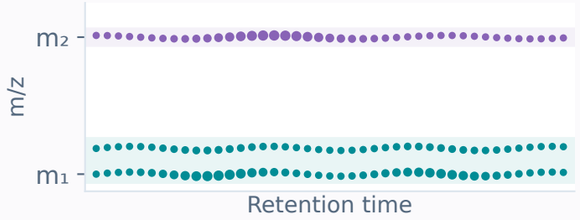
Synthetic centroid graph • actual Peak3D links and basin labels • no fitted surface or voxel grid

The difference is how peak candidates are formed

Both approaches start with retention time, m/z and intensity. They organise the data differently.

TRACE-BASED DETECTION

1 Assemble traces / ROIs



2 Find peaks within traces



3 Report peak candidates

A trace is the parent object. Chromatographic segmentation defines peaks inside it.

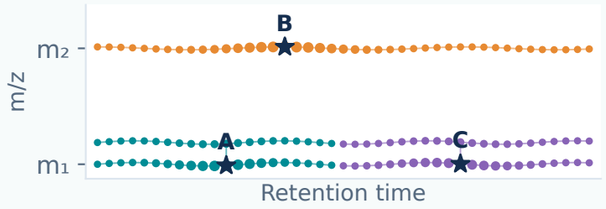
Trace building may use intensity and adaptive tolerances; implementations differ.

Peak3D CANDIDATE DETECTION

1 Link local centroids uphill



2 Form and merge apex basins



3 Validate peak candidates

A basin is the candidate object. Apply basin gates, an EIC check, and the final score.

Dot size shows intensity. Peak3D panels show actual graph and basin assignments.

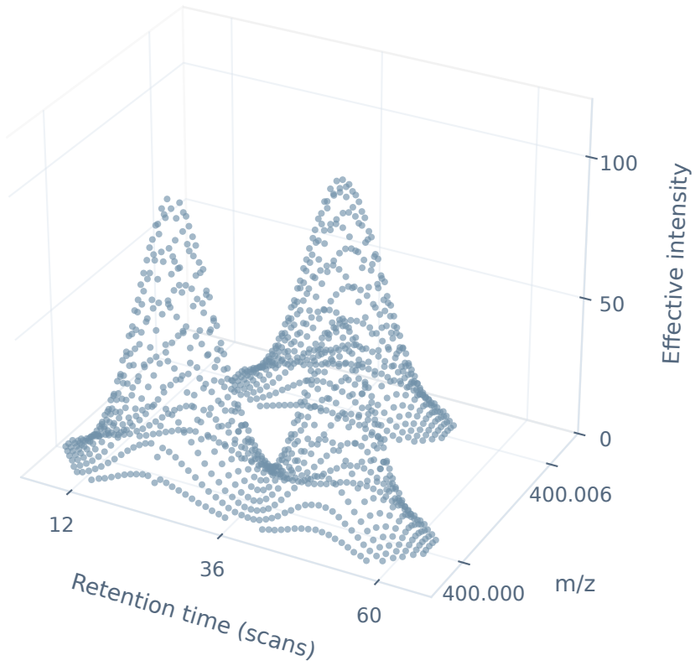
Trace-based: form traces, then find peaks. Peak3D: form apex basins directly from centroids.

Architecture schematic • centWave / MassCube: ROI then segmentation; asari: composite mass tracks • sources in notes

A basin is a cloud of points that leads to one apex

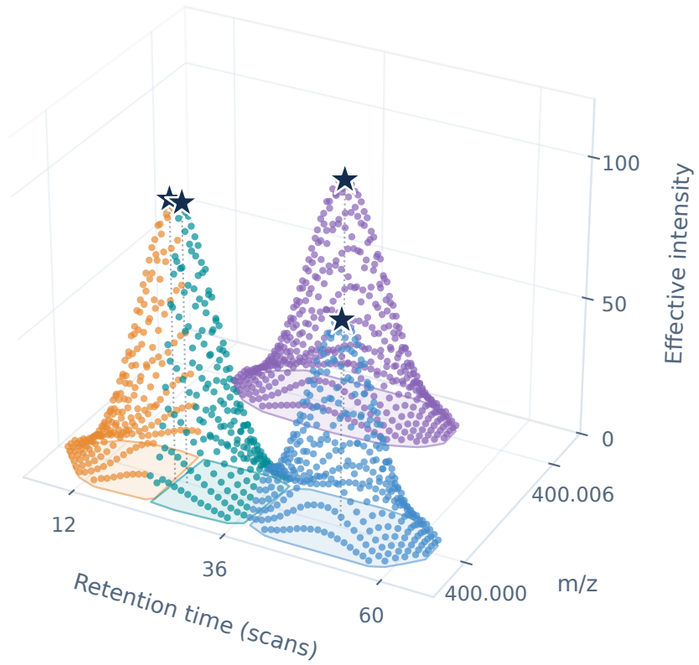
Each dot is a centroid. Colour shows basin membership; a star marks its apex.

1 Start with a point cloud



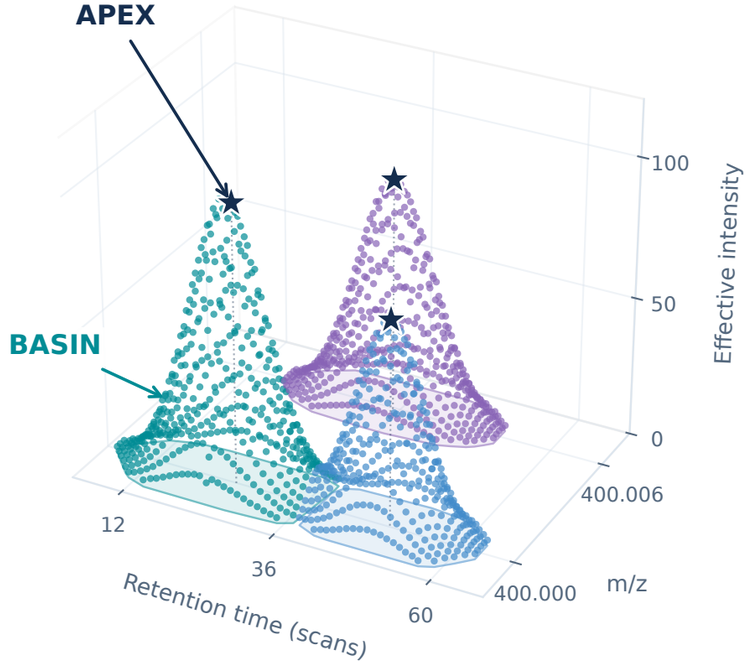
Positions: retention time and m/z.
Height: effective intensity.

2 Assign points to apexes



Four initial basins.
The shallow dip creates an extra apex.

3 Merge the shallow split



Three surviving basins.
The weak split joins its neighbour.

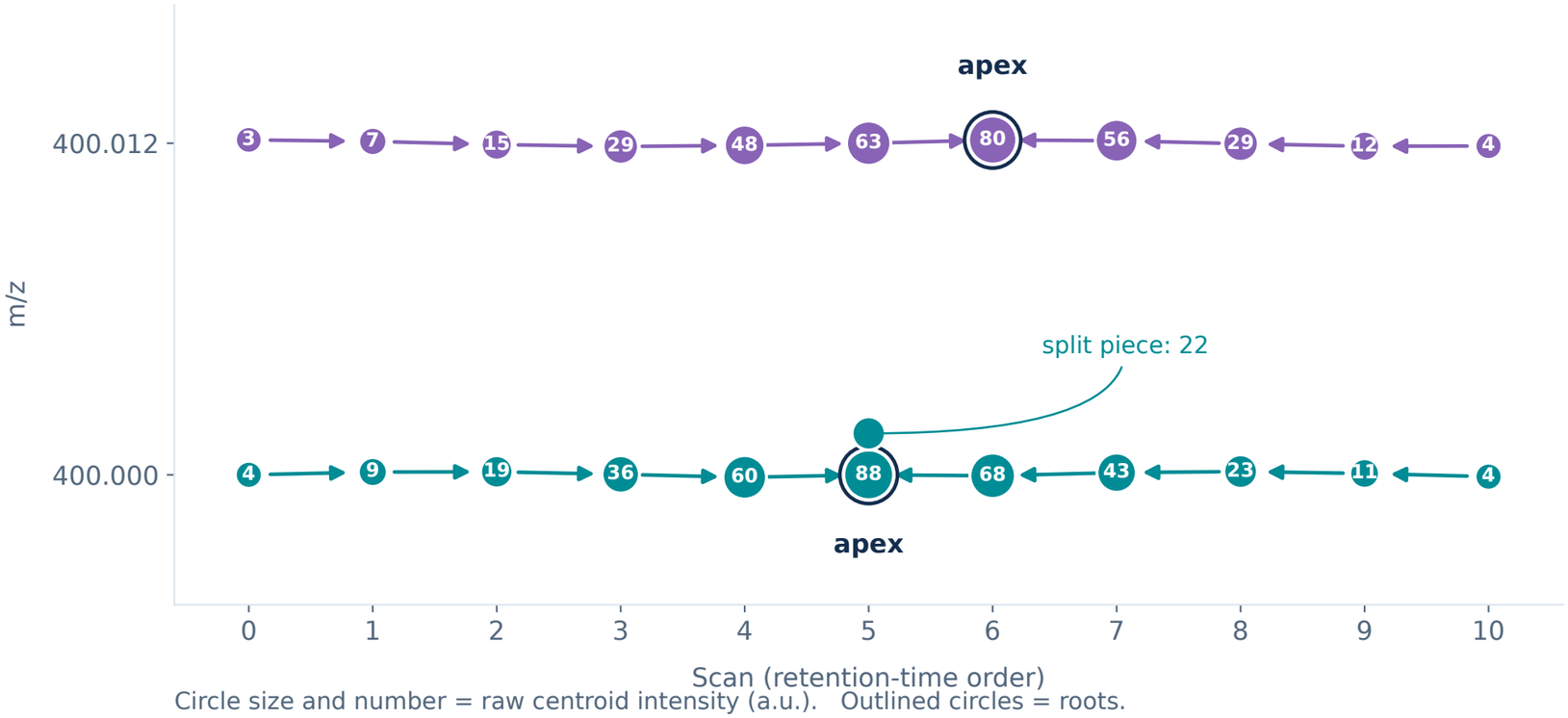
Same colour = one basin • ★ = apex • Very low-intensity points omitted for clarity

BASIN = points sharing an apex. **APEX = the endpoint of their uphill paths.**

Illustrative synthetic clouds with broadened m/z spread • memberships computed by Peak3D • faint patches = projected footprints

Each centroid follows a path to one apex

Arrows show the actual parent links. This is a top-down view of the point cloud.



Circle size and number = raw centroid intensity (a.u.). Outlined circles = roots.

01 Find eligible neighbours

Use the pair's m/z tolerance in nearby scans; link same-scan split centroids too.

02 Choose the highest neighbour

Move to a higher effective intensity.
If none is higher, this point is an apex.

03 Follow the arrows

All points reaching the same root belong to one initial basin.

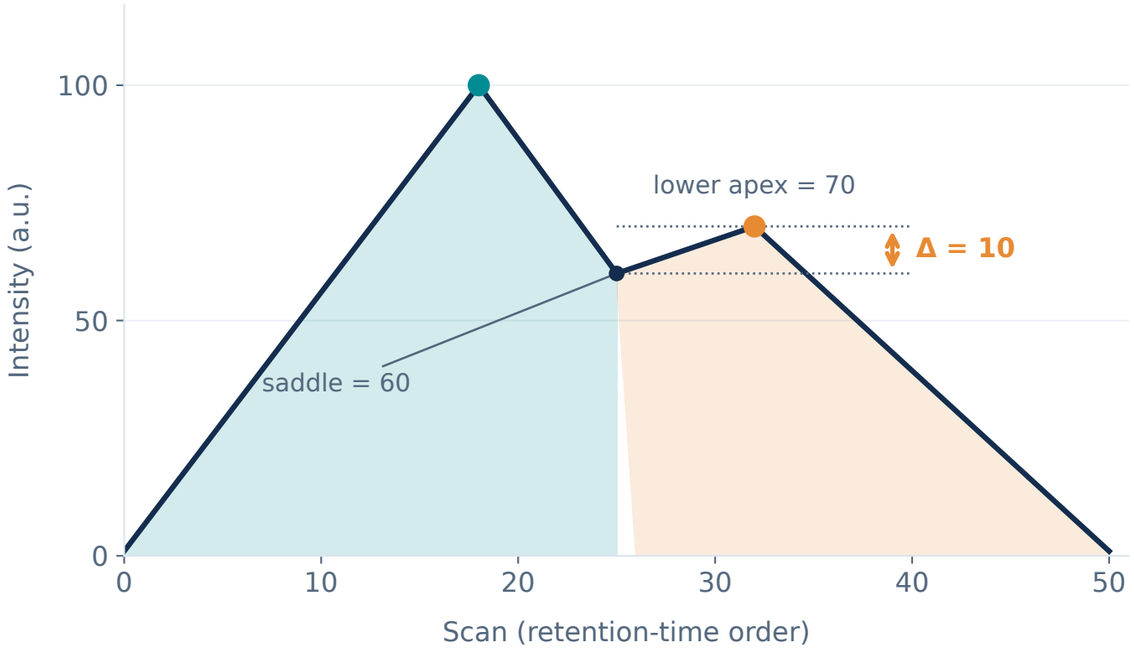
Split example: 88 + 22 = 110
effective height; the stronger piece is the root.

A basin is the set of centroids whose uphill paths end at the same apex.

Should two neighbouring basins be one peak?

Persistence measures how far the lower apex rises above the connecting valley (saddle).

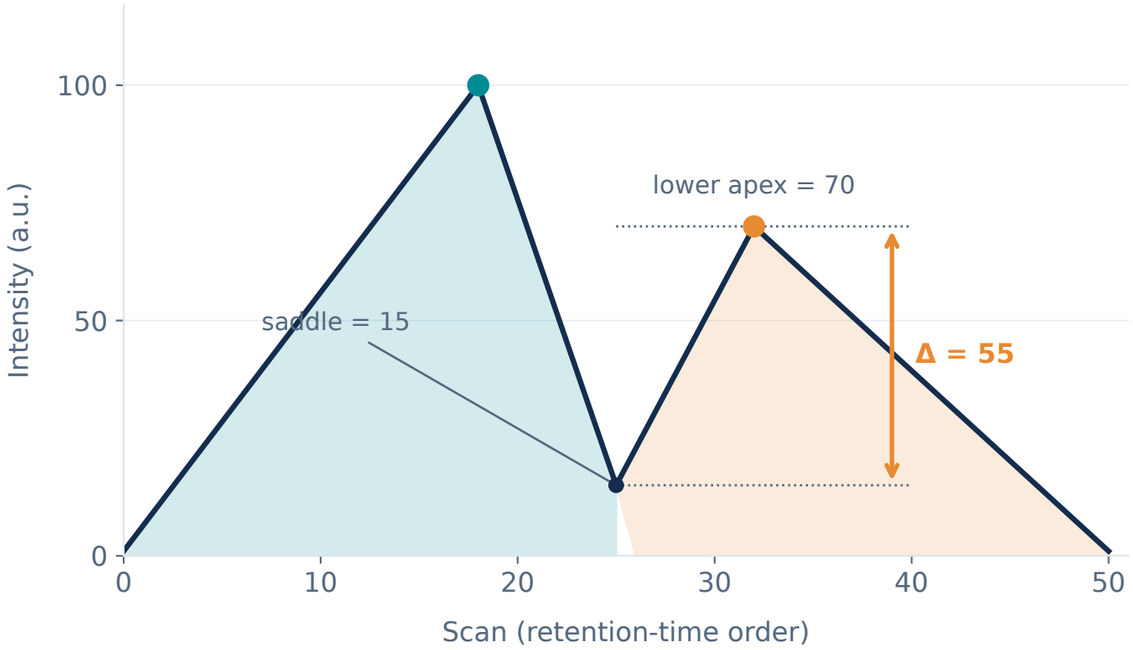
Shallow dip



MERGE → one candidate

Δ = 10 < threshold 35

Deep valley



RETAIN → two candidates

Δ = 55 ≥ threshold 35

Here: threshold = max(0.5 × lower apex, 3 × local noise) = 35

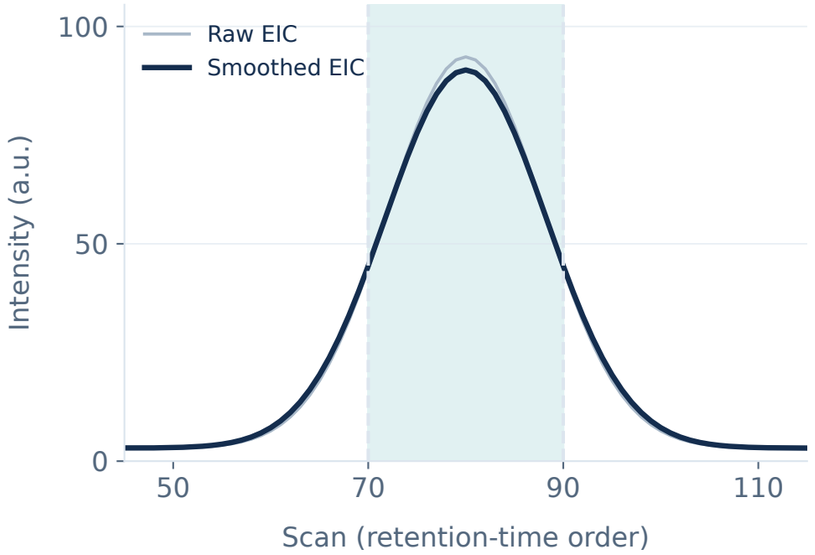
In these examples, a shallow dip merges; a deep, noise-resolved valley keeps two peaks.

Synthetic worked examples • default fractional merge rule • spacing and noise also affect the decision

A 3D candidate must also look like a chromatographic peak

After basin-level gates, inspect the extracted ion chromatogram (EIC) around each candidate.

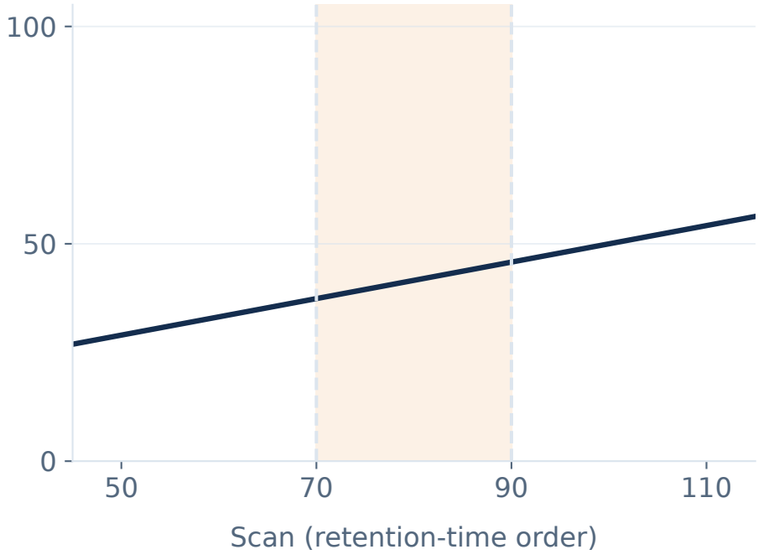
Localised peak



PASS chromatogram check

A maximum lies inside the bounds and rises clearly above the noise.

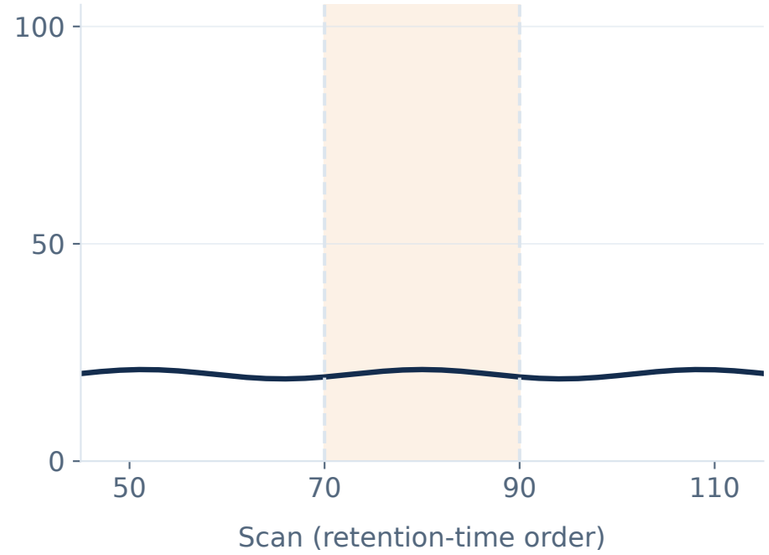
Fragment of a rising signal



REJECT: no local apex

The signal keeps rising through the candidate's entire interval.

Weak ripple on background



REJECT: too little prominence

A maximum exists, but its prominence is below 3 × the noise reference.

Shaded interval = candidate bounds. Other checks cover scan support, width, m/z consistency, background and score.

3D basins propose candidates; chromatogram checks help reject fragments and noise.

Synthetic candidates passed to the actual EIC check • these are check outcomes, not final feature calls